

① Teste 7

$$f_0 = \frac{v + v_0}{v + v_s} f_s$$

$$\frac{v_0}{v_s} = \frac{15}{35}$$

$$\frac{v_0}{v_s} = \frac{15,0 \text{ m/s}}{35 \text{ m/s}}$$

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$$f = 392 \text{ Hz}$$

$$v_A = 0$$

$$v_0 = 15 \text{ m/s}$$

$$v_B = 35 \text{ m/s}$$

Velocidade do som = $340,29 \text{ m/s}$

$$a) f_0^A = \frac{v - v_0}{v + v_A} f = \frac{340,29 \text{ m/s} - 15 \text{ m/s}}{340,29 \text{ m/s}} (392 \text{ Hz})$$

$$= 374,72 \text{ Hz} //$$

$$b) f_0^B = \frac{v + v_0}{v + v_B} f = \frac{(340,29 + 15) \text{ m/s}}{(340,29 + 35) \text{ m/s}} (392 \text{ Hz})$$

$$= 371,109 \text{ Hz} = 371,109 \text{ Hz}$$

~~Considerando a velocidade de referência.~~ ~~v_0 como referência.~~

~~$$v_B' = v_B - v_0 = 35 - 15 = 20 \text{ m/s}$$~~

~~$$f_0^B = \frac{v' + 0}{v' + 20} f = \frac{(340,29 + 15) \text{ m/s}}{(340,29 + 15 - 20) \text{ m/s}} f$$~~

4) c)

$$f_{\text{Bat}} = |f'_A - f'_B|$$

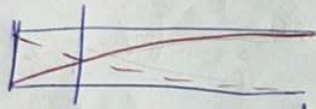
$$= 374,9 - 371,3 = 3,642$$

② $F_4 (F_0)$ e) $T_i = 20^\circ\text{C}$

$$C \Rightarrow f = 349 \text{ Hz}$$

$$v = 343,09 \frac{\text{m}}{\text{s}}$$

$$F_4 = 344 \text{ Hz}$$



$$l = \frac{\lambda}{4} \text{ fundamental}$$

$$v = \lambda f$$

$$\lambda = \frac{v}{f} \Rightarrow l = \frac{v}{4f}$$

$$l = \frac{343 \text{ m/s}}{4(349 \text{ Hz})}$$

$$= \frac{343}{1396} = 0,2457 \text{ m}$$

b) 370 Hz no $F \rightarrow F\#$

Qual seria a temperatura?

$$v \approx 331 + 0,6T$$

$$\frac{f_2}{f_1} = \frac{v_2}{v_1}$$

$$\frac{370 \text{ Hz}}{349 \text{ Hz}} = \frac{v_2}{343 \frac{\text{m}}{\text{s}}}$$

$$v_2 = 363,64 \frac{\text{m}}{\text{s}}$$

$$v = 331 + 0,6T$$

$$T = \frac{v - 331}{0,6}$$

$$T = 54,4^\circ\text{C}$$

③ $\lambda = 64,8 \text{ cm} = 64,8 \times 10^{-2} \text{ m}$
 $\lambda' = 65,2 \text{ cm} = 65,2 \times 10^{-2} \text{ m}$

$$V = \lambda f \quad f = \frac{V}{\lambda}$$

$$f_{\text{bat}} = |f_1 - f_2| = \left| \frac{V}{\lambda} - \frac{V}{\lambda'} \right|$$

$$= \left| \frac{343 \text{ m/s}}{0,648 \text{ m}} - \frac{343 \text{ m/s}}{0,652 \text{ m}} \right| = 3,25 \text{ Hz}$$

$$f_1 = \frac{V}{\lambda} = \frac{343 \text{ m/s}}{0,648 \text{ m}} = 529,32 \text{ Hz}$$

$$f_2 = \frac{V}{\lambda'} = \frac{343 \text{ m/s}}{0,652 \text{ m}} = 526,07 \text{ Hz}$$

$$(529,32 - 526,07) \text{ Hz} = 3,25 \text{ Hz}$$

$$343 \frac{\text{m}}{\text{s}} \left(\frac{1}{0,648} - \frac{1}{0,652} \right) \frac{1}{\text{m}} = 3,247 \text{ Hz}$$

④  $\left. \begin{array}{l} 0,62 \text{ m} \\ m = 0,00725 \text{ Kg} \\ 7,25 \times 10^{-3} \text{ Kg} \\ T = 4,110 \text{ N} \end{array} \right\}$

$$V = \sqrt{\frac{T}{\mu}} \quad \mu = \frac{m}{l} = \frac{7,25 \times 10^{-3} \text{ Kg}}{0,62 \text{ m}} = 0,012 \frac{\text{Kg}}{\text{m}}$$

$$V = \sqrt{\frac{4,110 \text{ N}}{0,012 \frac{\text{Kg}}{\text{m}}}} = 592,15 \frac{\text{m}}{\text{s}} \quad f_s = \frac{V_c}{2l} = \frac{592,15 \frac{\text{m}}{\text{s}}}{2 \cdot 0,62 \text{ m}}$$

$$f_s = 478,1 \text{ Hz}$$

segundo sobreton $\div 3$ ton

$$n=3 \Rightarrow f_3 = 3f_s$$

$$f_3 = 3(478,5) \text{ Hz}$$

$$= 1434,324 \text{ Hz}$$

Para que o tubo tenha vibração em freq. fundamental!

$$f = f_s$$

$$v = \lambda f$$

$$f = \frac{v}{\lambda} = \frac{v}{4l}$$

$$1434,324 \text{ Hz}$$

$$\frac{343 \text{ m/s}}{4l} = 1434,324 \text{ Hz}$$

$$4l$$

$$l = 0,05957 \text{ m}$$

